the car, Slim streaks down the mile-long strip at 120 miles per hour while Jim stands on the sideline and times him. Wanting an independent confirmation, Slim also uses a stopwatch to determine how long it takes his new car to traverse the track. Prior to Einstein's work, no one would have questioned that if both Slim and Jim have properly functioning stopwatches, each will measure the identical elapsed time. But according to special relativity, while Jim will measure an elapsed time of 30 seconds, Slim's stopwatch will record an elapsed time of 29.99999999999952 seconds—a tiny bit less. Of course, this difference is so small that it could be detected only through a measurement whose accuracy is well beyond the capacity of hand-held stopwatches run by the press of a finger, Olympic-quality timing systems, or even the most precisely engineered atomic clocks. It is no wonder that our everyday experiences do not reveal the fact that the passage of time depends upon our state of motion.

There will be a similar disagreement on measurements of length. For example, on another test run Jim uses a clever trick to measure the length of Slim's new car: he starts his stopwatch just as the front of the car reaches him and he stops it just as the back of the car passes. Since Jim knows that Slim is speeding along at 120 miles per hour, he is able to figure out the length of the car by multiplying this speed by the elapsed time on his stopwatch. Again, prior to Einstein, no one would have questioned that the length Jim measures in this indirect way would agree exactly with the length Slim carefully measured when the car sat motionless on the showroom floor. Special relativity proclaims, on the contrary, that if Slim and Jim carry out precise measurements in this manner and Slim finds the car to be, say, exactly 16 feet long, then Jim's measurement will find the car to be 15.99999999999974 feet long—a tiny bit less. As with the measurement of time, this is such a minuscule difference that ordinary instruments are just not accurate enough to detect it.

Although the differences are extremely small, they show a fatal flaw in the commonly held conception of universal and immutable space and time. As the relative velocity of individuals such as Slim and Jim gets larger, this flaw becomes increasingly apparent. To achieve noticeable differences, the speeds involved must be a sizeable fraction of the maximum possible speed—that of light—which Maxwell's